CHIHYEONG LEE
chi0412@snu.ac.kr
Advisor: **Dr. Jooeun Ahn**

Looking for:
POSTDOC. POSITION

Evaluating, Understanding, and Predicting Human Movement

SPORTS ENGINEERING LAB

EDUCATION

Seoul National University, Sports Engineering Lab
- Ph.D. in Sports Science (***Expected Feb 2027)***
- M.S. in Sports Science, 2023

University of Houston, Health and Human Performance
- Visiting Researcher, 2023

RESEARCH PHILOSOPHY

"My goal is to integrate biomechanics, wearable robotics, and digital human simulation to evaluate, understand and predict human movement."

PUBLISH

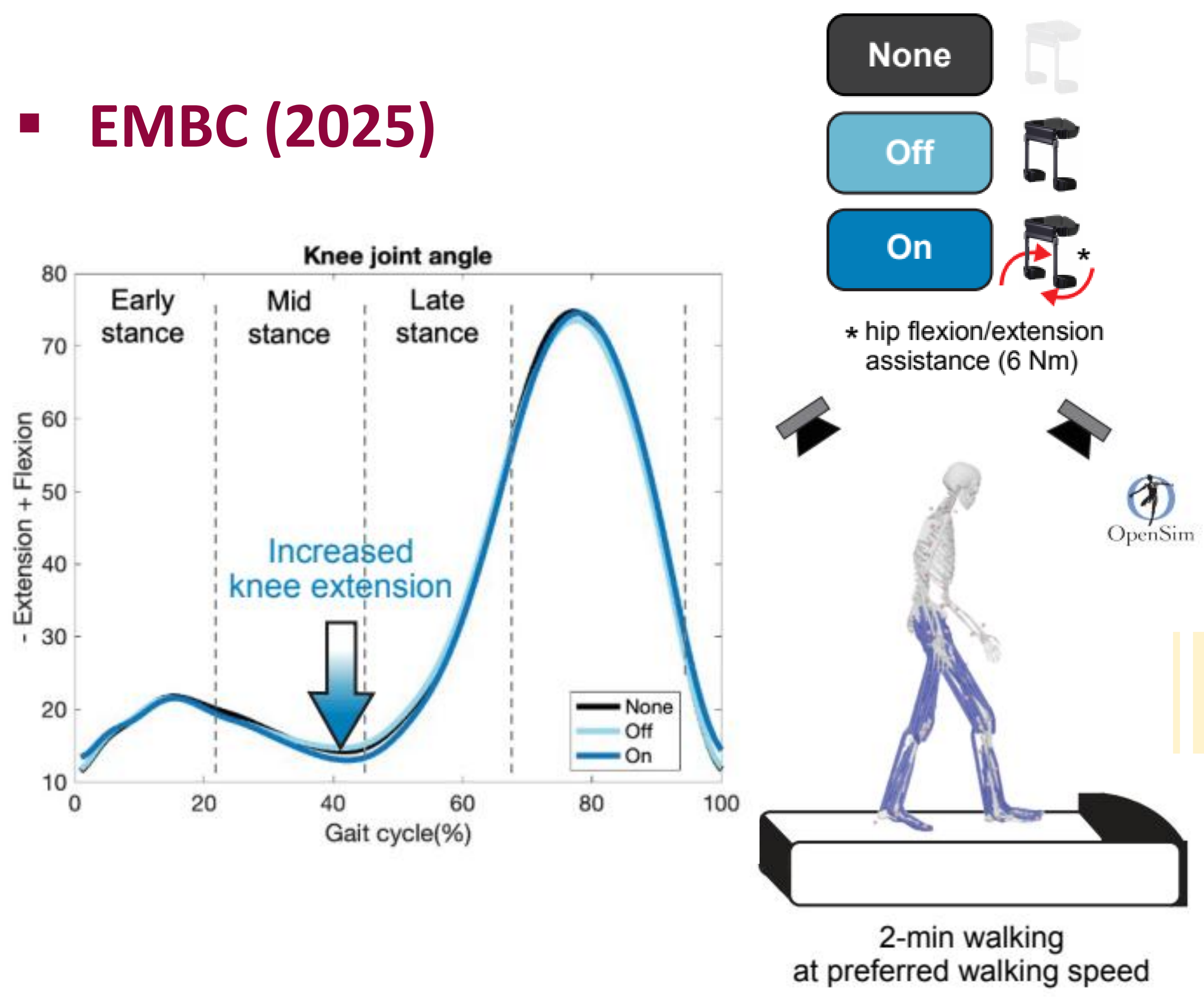
- **PAPERS:** 3 papers (3 first-author) + 1 under review
- **CONFERENCES:** 6 abstracts, 3 full papers

HUMAN-ROBOT INTERACTION

- **RAL (2025)**

"The proposed fully ***soft lift-assist wearable suit*** *showed the potential to* ***reduce muscle activation and metabolic cost during lifting tasks****."*

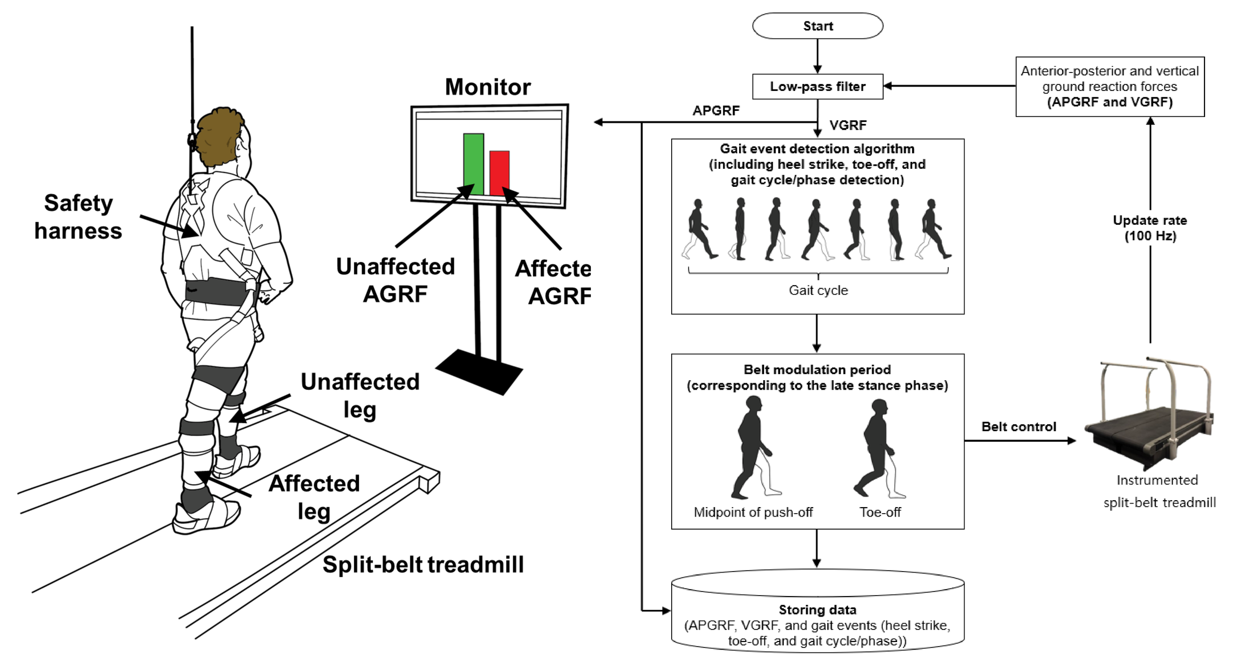
- **EMBC (2025)**



"Vector coding technique revealed that a ***hip assist robot altered hip-knee coordination****, particularly during the mid-to-late stance phase."*

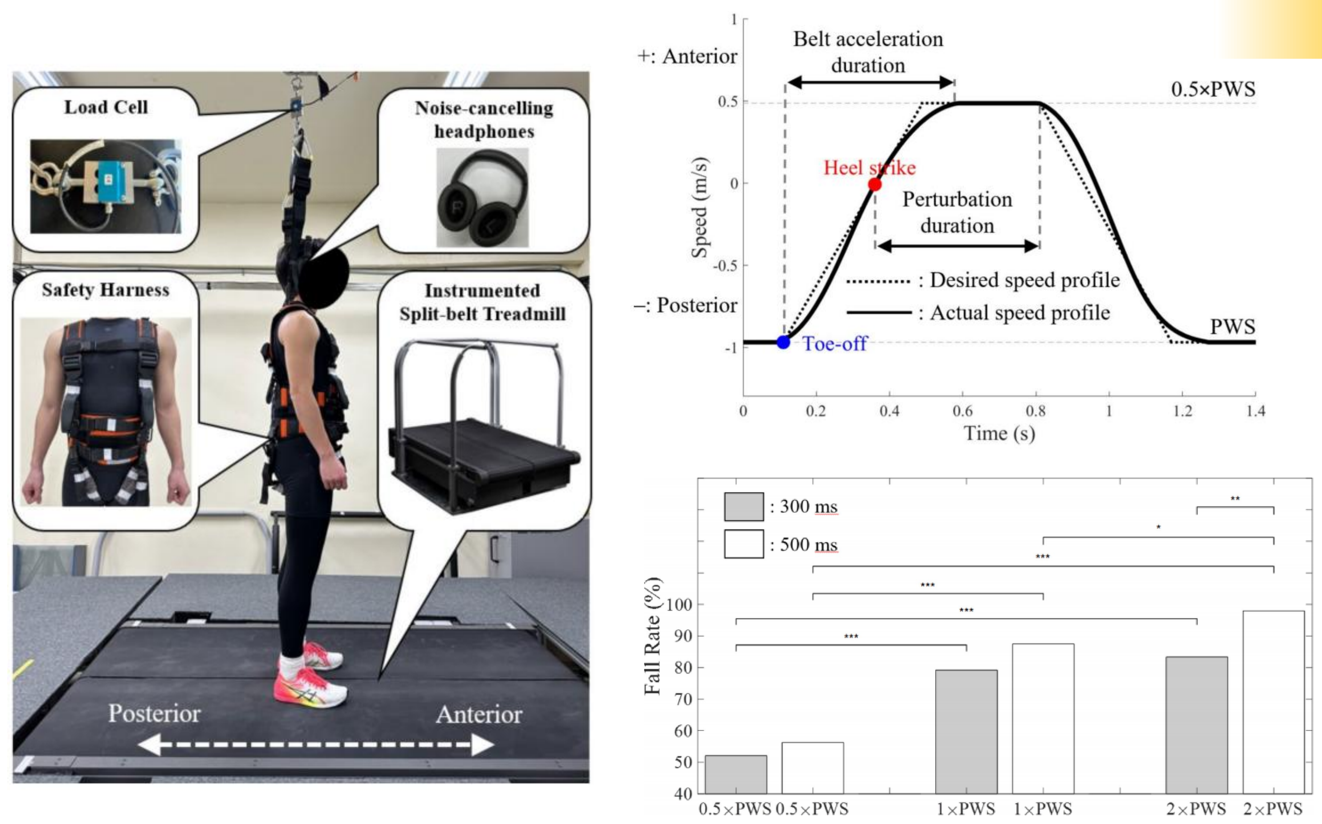
HUMAN RESPONSE ANALYSIS

- **JNER (2026, under review)**



"We developed ***a real-time post-stroke gait training system by customizing*** *an instrumented split-belt treadmill.* ***Visual biofeedback and adaptive belt deceleration synergistically enhanced gait propulsion.****"*

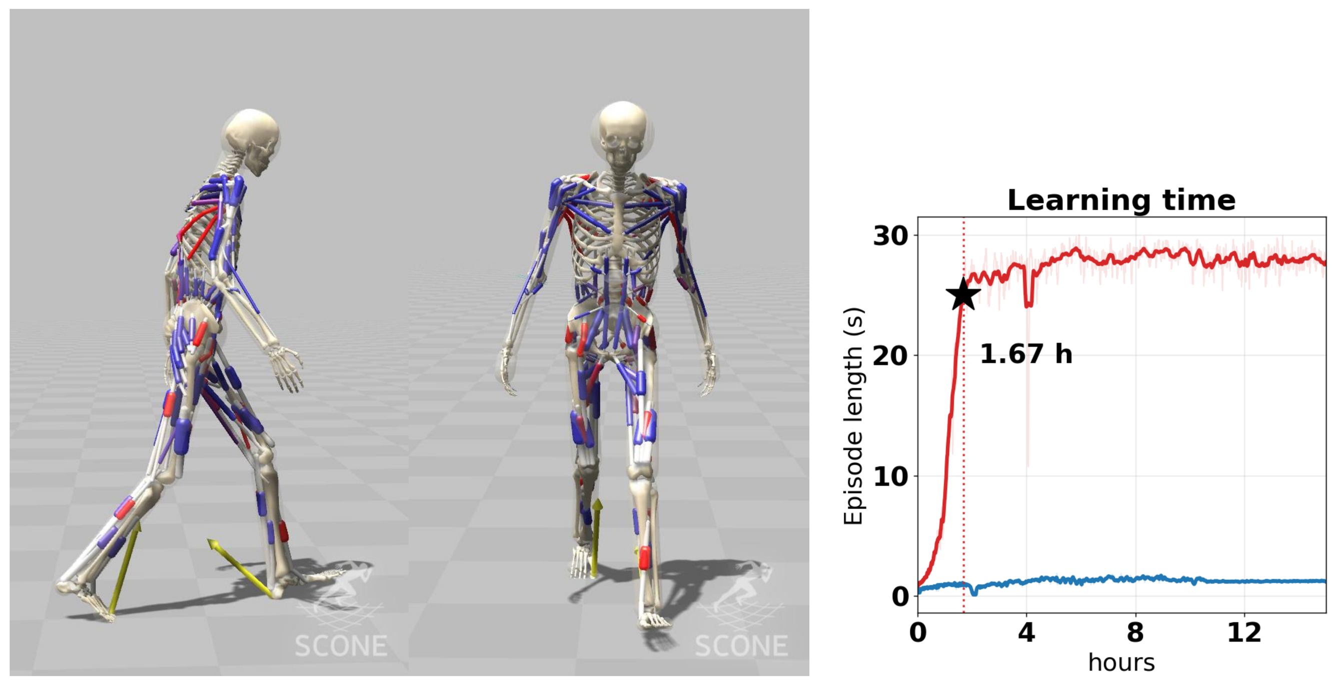
- **Scientific Reports (2025)**



"We customized an instrumented split-belt treadmill to ***deliver real-time slip perturbations*** *and* ***establish design criteria for fall-inducing experiments.****"*

HUMAN MOVEMENT PREDICTION

- **WCB (2026)**

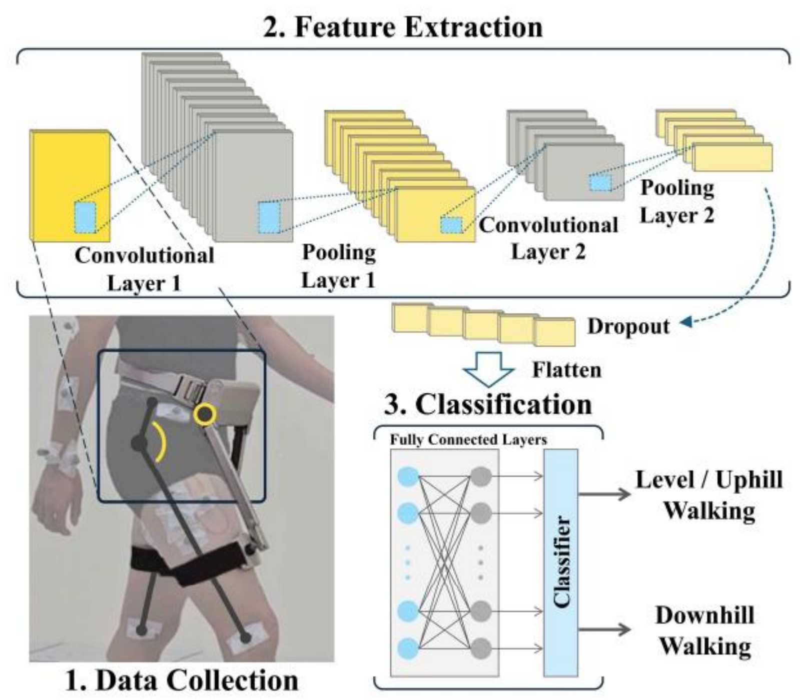


*"****Reference Configuration Imitation Learning (RCIL)*** *achieved fast and stable walking by learning residual posture corrections rather than direct muscle activations."*

- **ISBS (2023)**

*"****Machine learning*** *enabled accurate and* ***real-time detection of backward slips*** *for safer gait-assistive wearable robots."*

- **ASB (2025)**



"A ***deep learning*** *approach enabled* ***accurate and real-time classification of downhill walking*** *for adaptive gait-assistive wearable robots."*

SKILLS

Experimental Device

OptiTrack
AMTI FORCE AND MOTION
BERTEC
QUALISYS
DELSYS

+ Custom Software

Programming

Musculoskeletal Simulation

Visual 3D
OpenSim
HYFYDY HIGH FIDELITY DYNAMICS

LET'S STAY IN TOUCH!

LinkedIn
GitHub Pages
Google Scholar
SPORTS ENGINEERING LAB